Unit 6, Lesson 10: Solving Multi-Step Proportion Problems

Benchmark

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

Learning Target

- ☐ I can solve multi-step problems involving proportions.

6.10.1 Warm-Up: Making Maps

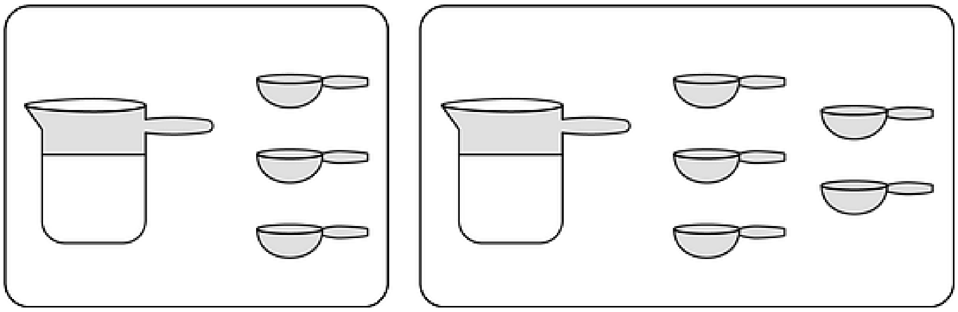
Cartographers and photogrammetrists are the professionals who collect and analyze data that they then use to create maps. Cartographers design accessible maps for general use, while photogrammetrists create more detailed maps of Earth's features.

1. What are some of the ways mathematics can be used for map creation?

6.10.2 Collaboration: The "Right" Way

Work with your partner to answer the following questions.

Lucas's favorite hot cocoa recipe calls for 1 cup (c) of milk and 3 tablespoons (tbsp) of cocoa. Today, he accidentally added 5 tablespoons of cocoa to 1 cup of milk and it was way too chocolatey.



1. His sister Meredith told him he could just add more milk to fix the flavor. How much milk would he need to add to get the right flavor?

2. His brother Tyler told him he could fix it by adding both milk and cocoa. How much more of each would he need to add to get the right flavor?

3. Which sibling had the correct solution to Lucas's problem? Explain.

6.10.3 Activity: Gallery Walk

With your group, read the information given in each scenario. Then, write three different questions that could be answered using the information that was given.

Scenario	Questions That Could Be Answered
Two Maps	
Favorite Pasta	
Making Muffins	

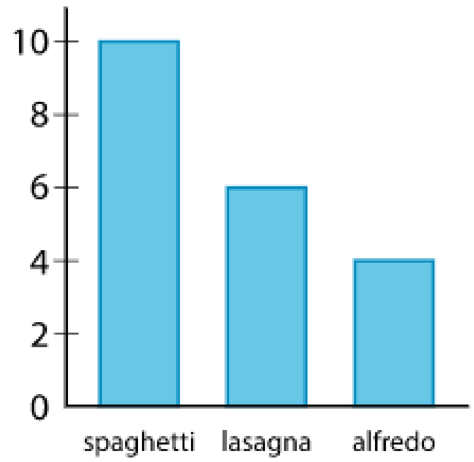
Choose one scenario. Discuss how each question could be answered. Then complete the table.

Question	My Summary of How to Answer the Question

6.10.4 Your Turn!

1. Chen surveys a random sample of students about their favorite pasta dish served by the cafeteria and makes a bar graph of the results.

   There are $1{,}256$ students who attend the school. Use Chen's data to estimate the number of students at the school whose favorite dish is lasagna.

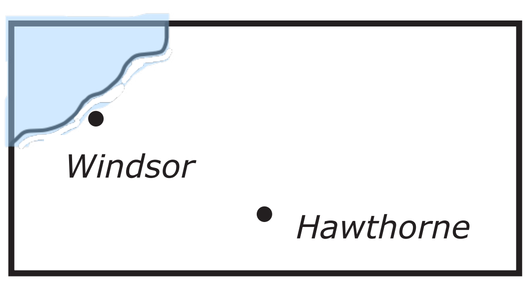
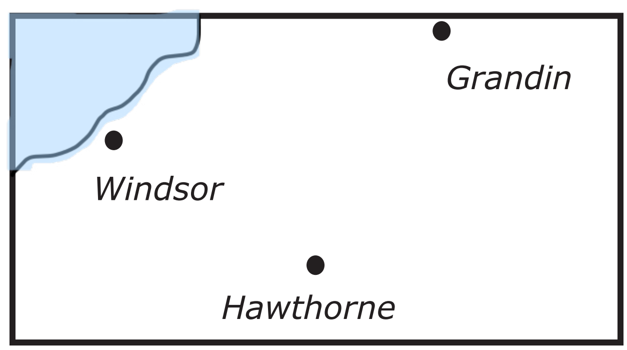


2. Jaxon is baking blueberry muffins with a recipe that calls for 2 cups (c) of flour and $\frac{3}{4}$ cup of sugar to make 12 muffins. Jaxon wants to make as many muffins as possible using 5 cups of flour.

   a. Exactly how many cups of sugar will Jaxon use if he uses all 5 cups of flour?

   b. Exactly how many muffins will Jaxon make if he uses all 5 cups of flour?

3. The two maps show the same area at two different scales.

Map A	**Map B**
 $1\ cm = 3.5\ mi$	

Grandin is not on Map A, and Map B does not have a scale.

- Windsor and Hawthorne are 2.9 cm apart on Map A.
- Windsor and Hawthorne are 4.3 cm apart on Map B.
- Windsor and Grandin are 6.2 cm apart on Map B.

What is the distance, in miles, between Windsor and Grandin? Round to the nearest hundredth.

6.10.5 Wrap-Up: Ticket Out the Door

1. David's car can travel about 25 miles per gallon and his fuel tank holds 12 gallons. He's driving 663 miles on a road trip from Miami, Florida to Atlanta, Georgia. If he starts with a full tank of gas in Miami, how many times will he have to stop for gas on the trip to Atlanta?